

Question 0**[8 points]**

For each of the following scenarios, choose the "best" data structure or a combination of data structures from the following list: an unsorted array, linked list, doubly linked list, circular linked list, stack. In each case justify your answer briefly.

- a) A list must be maintained so that any element can be accessed randomly in constant amount of time
- b) A program needs to remember operations it performed in opposite order.
- c) The size of a file is unknown. The entries need to be entered as they come in. Entries must be deleted when they are no longer needed. It is important that structure has flexible memory management.
- d) A list must be maintained so that elements can be added/deleted to the beginning or end in $\Theta(1)$ time

Question 1**[7 points]**

Suppose you were asked to write a method in C++ that will take two sorted stacks A and B (min on top of each) and create one stack that is sorted (min on top). You are allowed to use only the stack operations such as pop, push, size and top. No other data structure such as arrays are allowed. You are allowed to use stacks.

Question 2**[15 points]**

Assume that you have a doubly linked list with the head node and at least one other internal node M which is not the last node. Write few lines of code in C++ to accomplish the following. You may assume that each node has a **next** pointer and **prev** pointer. You may NOT swap data to accomplish any of the following operations. For each operation, assume the original list as described above. You are encouraged to draw pictures to justify your code. Note that for each operation, you need to manipulate at least two pointers, next and prev.

- a) Delete the head node.
- b) Insert a node P immediately after M.
- c) Swap head and the node M (you may not swap the data).

Question 3**[10 points]**

Perform time complexity analysis for following pieces of code, expressing the runtime in Θ notation.

- a)

```
int count=0;
for(int i=0; i<n; i+=2)
    for(int j=1; j<n; j++)
        count++;
```
- b)

```
int count=0;
for(int j=1; j<n; j+=2)
    for(int i=1; i<n; i++)
        count++;
```